Viral-load-informed outbreak risk following isolation

A concise methodology note

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Purpose. This note develops a branching-process method that converts viral-load trajectories into probabilities of a major outbreak. Infectors may differ by vaccination status and isolation behaviour. The same framework quantifies the residual outbreak risk caused by transmission after release from isolation.

1 Overview and assumptions

The calculation proceeds in four stages:

1. convert each viral-load trajectory into a time-varying infectiousness profile;
2. integrate that profile to measure the source's transmission potential;
3. adjust for recipient susceptibility and for the reduction in transmission during isolation; and
4. use the multitype branching-process result of Hart et al. to calculate the probability of a major outbreak [1].

Throughout, τ denotes time since infection. The label $s \in \{u, v\}$, used as either a subscript or superscript, records the infected individual's vaccination status: u for unvaccinated and v for vaccinated.

The analysis describes the early phase of spread as a branching process. It therefore neglects susceptible depletion, treats transmission lineages as independent, and assumes that each infected individual generates infections through an inhomogeneous Poisson process. Contacts are random, and vaccination affects acquisition through a relative-susceptibility factor.

2 From viral load to transmission potential

An infected individual with vaccination status s has viral-load trajectory $V_s(\tau)$. We write the corresponding instantaneous infectiousness as

$$\beta_s(\tau) = f(V_s(\tau))\,, \tag{2.1}$$

where f maps viral load to infectiousness. A Hill function is one possible choice, but the methodology does not depend on a particular functional form. The viral-load model and infectiousness curve used for numerical implementation are specified in section 8.

Integrating the infectiousness profile over the full infection gives

$$B_s = \int_0^\infty \beta_s(\tau)\,\mathrm{d}\tau. \tag{2.2}$$

The quantity B_s can be interpreted as the expected number of transmissions from a source of type s when all contacted recipients have unit susceptibility.

2.1 Recipient susceptibility

Let θ be the vaccinated proportion of the population. Vaccinated recipients have relative susceptibility η, whereas unvaccinated recipients have relative susceptibility 1. Under random mixing, the mean relative susceptibility of a contacted recipient is

$$\overline{\eta} = \theta\eta + (1-\theta). \tag{2.3}$$

Multiplying the source's transmission potential by this mean susceptibility gives the expected number of infections in the partly vaccinated population:

$$R_s = \overline{\eta} B_s. \tag{2.4}$$

This distinction is useful: B_s describes the infectiousness of the source, whereas R_s also incorporates the susceptibility composition of the recipient population.

The branching-process calculation also requires the vaccination-status distribution among newly acquired local infections. Let a_s be the probability that such an infection occurs in a recipient with status s. Because vaccinated and unvaccinated recipients have different susceptibilities, this distribution need not match the population split. Under random mixing, each population share is weighted by its relative susceptibility and then normalised. Thus

$$a_v = \frac{\theta\eta}{\overline{\eta}}, \qquad a_u = \frac{1-\theta}{\overline{\eta}}. \tag{2.5}$$

3 Accounting for isolation

Consider a source with vaccination status s, detected at time $\tau_s^{(\mathrm{det})}$ and isolated for a duration T_s. The release time is

$$\tau_s^{(\mathrm{rel})} = \tau_s^{(\mathrm{det})} + T_s. \tag{3.1}$$

During isolation, suppose infectiousness is scaled by $\varepsilon \in [0,1]$ relative to infectiousness without isolation. Thus $\varepsilon = 0$ represents perfect isolation, while $\varepsilon = 1$ represents no reduction in infectiousness. Splitting the full infectious period into the intervals before detection, during isolation, and after release gives

$$B_s^{(\mathrm{iso})} = \int_0^{\tau_s^{(\mathrm{det})}} \beta_s(\tau)\,\mathrm{d}\tau + \varepsilon \int_{\tau_s^{(\mathrm{det})}}^{\tau_s^{(\mathrm{rel})}} \beta_s(\tau)\,\mathrm{d}\tau + \int_{\tau_s^{(\mathrm{rel})}}^{\infty} \beta_s(\tau)\,\mathrm{d}\tau. \tag{3.2}$$

As before, multiplying by mean recipient susceptibility converts this transmission potential into an expected number of infections:

$$R_s^{(\mathrm{iso})} = \overline{\eta} B_s^{(\mathrm{iso})}. \tag{3.3}$$

For questions specifically about release from isolation, only the final interval is needed. The corresponding post-release quantities are

$$B_s^{(\mathrm{post})} = \int_{\tau_s^{(\mathrm{rel})}}^{\infty} \beta_s(\tau)\,\mathrm{d}\tau, \qquad R_s^{(\mathrm{post})} = \overline{\eta} B_s^{(\mathrm{post})}. \tag{3.4}$$

4 General outbreak-probability result

Different infection pathways can have different transmission potential because of factors such as vaccination status, isolation behaviour, and individual viral-load dynamics. Represent these pathways by mutually exclusive infection types $j = 1, \dots, n$. Let a_j be the probability that a typical newly acquired local infection has type j, with $\sum_{j=1}^{n} a_j = 1$, and let R_j be the expected number of infections generated by a source of that type.

Under the separable random-mixing and Poisson assumptions above, Hart et al. [1] show that the probability that one typical newly infected individual initiates a major outbreak satisfies

$$p_{\text{out}} = 1 - \sum_{j=1}^{n} a_j \exp(-R_j p_{\text{out}}) . \tag{4.1}$$

For a type-j source, $\exp(-R_j p_{\text{out}})$ is the probability that none of its secondary infections establishes a surviving transmission lineage. The sum averages this probability over the possible pathways.

This fixed-point equation always has the zero solution. The outbreak probability is the *largest* solution in $[0, 1]$, and it is positive precisely when

$$\mathcal{R}_{\text{eff}} := \sum_{j=1}^{n} a_j R_j > 1. \tag{4.2}$$

5 Vaccination and isolation as four infection types

Suppose the same proportion ϕ of vaccinated and unvaccinated infected individuals isolate. Combining the two vaccination statuses with the two isolation behaviours gives four infection types. Table 1 lists the probability that a newly acquired infection belongs to each type and the corresponding source-specific offspring mean.

Table 1: Infection types in the homogeneous-trajectory model.

Type j	New-infection probability a_j	Mean offspring R_j
Vaccinated, isolating	$\theta\eta\phi/\overline{\eta}$	$R_v^{(\text{iso})}$
Vaccinated, non-isolating	$\theta\eta(1-\phi)/\overline{\eta}$	R_v
Unvaccinated, isolating	$(1-\theta)\phi/\overline{\eta}$	$R_u^{(\text{iso})}$
Unvaccinated, non-isolating	$(1-\theta)(1-\phi)/\overline{\eta}$	R_u

Substituting the four rows of table 1 into equation (4.1) gives

$$\begin{aligned} p_{\text{out}} = 1 - \frac{1}{\overline{\eta}}\Big\{ &\theta\eta \left[\phi \exp\left(-R_v^{(\text{iso})} p_{\text{out}}\right) + (1-\phi)\exp(-R_v p_{\text{out}})\right] \\ &+ (1-\theta) \left[\phi \exp\left(-R_u^{(\text{iso})} p_{\text{out}}\right) + (1-\phi)\exp(-R_u p_{\text{out}})\right]\Big\}. \end{aligned} \tag{5.1}$$

The condition for a positive solution can likewise be written directly in terms of the transmission potentials:

$$\begin{aligned} \mathcal{R}_{\text{eff}} = \theta\eta &\left[\phi B_v^{(\text{iso})} + (1-\phi)B_v\right] \\ &+ (1-\theta) \left[\phi B_u^{(\text{iso})} + (1-\phi)B_u\right]. \end{aligned} \tag{5.2}$$

6 Residual risk after release from isolation

Equation (4.1) concerns a typical newly acquired local infection and therefore averages over its possible types. If the type of an index case is already known, no such averaging is needed for that source: its own offspring mean is used for the first generation. A known source of type j therefore initiates a major outbreak with probability

$$P_j = 1 - \exp(-R_j p_{\text{out}}), \tag{6.1}$$

where p_{out} still applies to each infection generated by the index case, because subsequent cases follow the usual type distribution.

The same argument applies when only a specified part of the infectious period is of interest. For an isolating source of known vaccination status s, the number of post-release transmissions is

$$N_s^{(\text{post})} \sim \text{Poisson}\Big(R_s^{(\text{post})}\Big). \tag{6.2}$$

The probability of at least one post-release transmission is therefore

$$\Pr\Big(N_s^{(\text{post})} \geq 1\Big) = 1 - \exp\Big(-R_s^{(\text{post})}\Big), \tag{6.3}$$

and the probability that at least one of these infections establishes a surviving transmission lineage, thereby producing a major outbreak, is

$$P_s^{(\text{post})} = 1 - \exp\Big(-R_s^{(\text{post})} p_{\text{out}}\Big). \tag{6.4}$$

If the whole infection is of interest instead, including transmission before detection, during isolation, and after release, equation (6.1) is used with $R_s^{(\text{iso})}$.

7 Heterogeneous viral-load trajectories and detection times

The homogeneous-trajectory assumption can be relaxed to allow variation in viral-load trajectories and/or detection times within each vaccination group. For status s, represent this joint variation by classes $i = 1, \ldots, m_s$. Class i has trajectory $V_{s,i}(\tau)$, detection time $\tau_{s,i}^{(\text{det})}$, and conditional probability $\pi_{s,i}$, where $\sum_i \pi_{s,i} = 1$. Its release time is $\tau_{s,i}^{(\text{rel})} = \tau_{s,i}^{(\text{det})} + T_s$, and it has its own $\beta_{s,i}$, $B_{s,i}$, $B_{s,i}^{(\text{iso})}$, $B_{s,i}^{(\text{post})}$, and corresponding R quantities.

These classes can be constructed in two natural ways:

- **A small set of representative classes.** Clustering or other machine-learning methods may identify a limited number of distinct viral-load patterns. Each class is assigned a representative trajectory, a detection time, and an estimated class probability $\pi_{s,i}$.
- **A large empirical sample.** Alternatively, many trajectory–detection-time pairs may be sampled from a fitted joint distribution. Treating each sampled pair as a class with weight $\pi_{s,i} = 1/m_s$ makes the finite sum an approximation to the integral over that distribution.

Using these class weights within each vaccination group, and retaining a common isolation probability ϕ, gives

$$\begin{aligned} p_{\text{out}} = 1 - \frac{1}{\eta}\Bigg\{ &\theta\eta \sum_{i=1}^{m_v} \pi_{v,i} \left[\phi\, \mathrm{e}^{-R_{v,i}^{(\text{iso})} p_{\text{out}}} + (1-\phi)\mathrm{e}^{-R_{v,i} p_{\text{out}}}\right] \\ &+ (1-\theta) \sum_{i=1}^{m_u} \pi_{u,i} \left[\phi\, \mathrm{e}^{-R_{u,i}^{(\text{iso})} p_{\text{out}}} + (1-\phi)\mathrm{e}^{-R_{u,i} p_{\text{out}}}\right]\Bigg\}. \end{aligned} \tag{7.1}$$

The corresponding average post-release outbreak probability among isolating sources of vaccination status s, accounting for variation in both trajectories and detection times, is

$$\overline{P}_s^{(\text{post})} = \sum_{i=1}^{m_s} \pi_{s,i} \left[1 - \exp\Big(-R_{s,i}^{(\text{post})} p_{\text{out}}\Big)\right]. \tag{7.2}$$

8 Ejima viral-load and infectiousness parameterisation

For numerical implementation, we generate viral-load trajectories using the target-cell-limited model fitted by Ejima et al. to symptomatic SARS-CoV-2 Delta cases in Singapore [2]. Their fitted time origin is symptom onset. To retain τ as time since infection, define the onset-centred time

$$t = \tau - \tau_s^{(\mathrm{sym})}, \tag{8.1}$$

where $\tau_s^{(\mathrm{sym})}$ is the time of symptom onset. Let $Q_s(t)$ be the number of uninfected target cells relative to its value at symptom onset, and let $\widetilde{V}_s(t)$ be viral load in copies per mL. The model is

$$\frac{\mathrm{d}Q_s(t)}{\mathrm{d}t} = -\xi_s Q_s(t)\widetilde{V}_s(t), \tag{8.2}$$

$$\frac{\mathrm{d}\widetilde{V}_s(t)}{\mathrm{d}t} = \gamma_s Q_s(t)\widetilde{V}_s(t) - \delta_s \widetilde{V}_s(t), \tag{8.3}$$

with

$$Q_s(0) = 1, \qquad \widetilde{V}_s(0) = V_{0,s}. \tag{8.4}$$

Here, ξ_s is Ejima et al.'s parameter β, renamed to avoid a clash with the infectiousness profile $\beta_s(\tau)$. The parameters γ_s and δ_s are the maximum viral-replication rate and the infected-cell loss rate, respectively. Solving from symptom onset both backward and forward gives

$$V_s(\tau) = \widetilde{V}_s\left(\tau - \tau_s^{(\mathrm{sym})}\right). \tag{8.5}$$

Vaccination status and between-individual variation enter through the model parameters. For $p \in \{\gamma, \delta, \xi, V_0\}$, the status-specific median is

$$\widetilde{p}_s = \begin{cases} p_0, & s = u, \\ p_0 \exp(c_{p,v}), & s = v, \end{cases} \tag{8.6}$$

where p_0 and $c_{p,v}$ are given in table 2. Individual parameters are sampled as

$$p_s = \widetilde{p}_s \exp(Z_{p,s}), \qquad Z_{p,s} \sim \mathcal{N}(0, \omega_p^2). \tag{8.7}$$

The supporting implementation samples the four random effects independently.

Viral load is converted to infectiousness using the Hill function

$$f(V) = \kappa \frac{V^\alpha}{V^\alpha + \lambda^\alpha}, \qquad \kappa = 0.546\ \mathrm{day}^{-1}, \quad \lambda = 10^7\ \mathrm{copies\,mL^{-1}}, \quad \alpha = 0.8. \tag{8.8}$$

Here, λ is the half-maximum viral load, α controls the slope, and κ sets the overall infectiousness scale. Ejima et al. calibrated this scale so that the mean reproduction number without isolation was 3.

Table 2: Viral-dynamics and infectiousness parameters adopted from Ejima et al. [2]. Parentheses contain standard errors.

Quantity and current notation	Unvaccinated baseline (SE)	Vaccination effect (SE)	Random-effect SD (SE)
Maximum viral-replication rate, γ (day^{-1})	4.25 (9.12)	0	3.05 (1.89)
Infected-cell loss rate, δ (day^{-1})	0.67 (0.04)	0.95 (0.14)	0.48 (0.05)
Target-cell infection rate, ξ [Ejima β] ($\mathrm{mL\,copy}^{-1}\mathrm{day}^{-1}$)	$8.03{\times}10^{-7}$ ($8.24{\times}10^{-7}$)	−1.81 (1.03)	0.77 (0.57)
Viral load at symptom onset, V_0 ($\mathrm{copies\,mL}^{-1}$)	$8.98{\times}10^{7}$ ($4.64{\times}10^{7}$)	1.26 (0.96)	2.29 (0.61)
Half-maximum viral load, λ ($\mathrm{copies\,mL}^{-1}$)	10^7	–	–
Hill slope, α	0.8	–	–
Overall infectiousness scale, κ (day^{-1})	0.546	–	–

Zero entries denote a fitted vaccination effect equal to zero, whereas dashes denote parameters fixed rather than estimated in the viral-dynamics model. The four viral-dynamics parameters are lognormally distributed, so ω_p is a standard deviation on the log scale. The fitted V_0 is implemented in copies per mL; logarithms are taken only when viral load is reported on a $\log_{10}$ scale. The standard errors quantify parameter-estimation uncertainty and are not included in the individual-level random-effect draws.

These estimates are specific to the symptomatic adult Delta cases used by Ejima et al.; applying the method to another variant or population would require re-estimation. Repeated parameter draws within each vaccination group generate the trajectory samples described in section 7.

9 Possible extensions

Several further extensions are straightforward:

- **Asymptomatic infections.** Symptomatic and asymptomatic infections can be treated as separate types, with different frequencies, infectiousness profiles, and detection or isolation behaviour.
- **Heterogeneous contact rates.** Source-side variation in contact activity can be incorporated by multiplying each class-specific infectiousness profile $\beta_{s,i}(\tau)$ by a contact-rate factor $c_{s,i}$. If contact activity also changes the chance of acquiring infection, the class weights must also be activity-weighted.
- **Non-random contact patterns.** Assortative contact between groups requires a full next-generation matrix. The scalar fixed point equation (4.1) is then replaced by the corresponding vector of extinction equations.
- **Age dependence.** Ejima et al. estimate an age effect on the target-cell infection rate ξ, which could be incorporated through age-specific parameter distributions or additional classes [2]. It may be useful to consider this alongside both age-dependent contact rates and non-random mixing between age groups.

References

[1] W. S. Hart, H. Park, Y. D. Jeong, K. S. Kim, R. Yoshimura, R. N. Thompson, and S. Iwami. Analysis of the risk and pre-emptive control of viral outbreaks accounting for within-host dynamics:

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[2] K. Ejima, M. Ajelli, A. Singh, et al. Age- and vaccination status-dependent isolation guidelines based on simulation of SARS-CoV-2 Delta cases in Singapore. *Communications Medicine* 5:76, 2025. doi:10.1038/s43856-025-00797-8. Supporting code: doi:10.5281/zenodo.10077030.